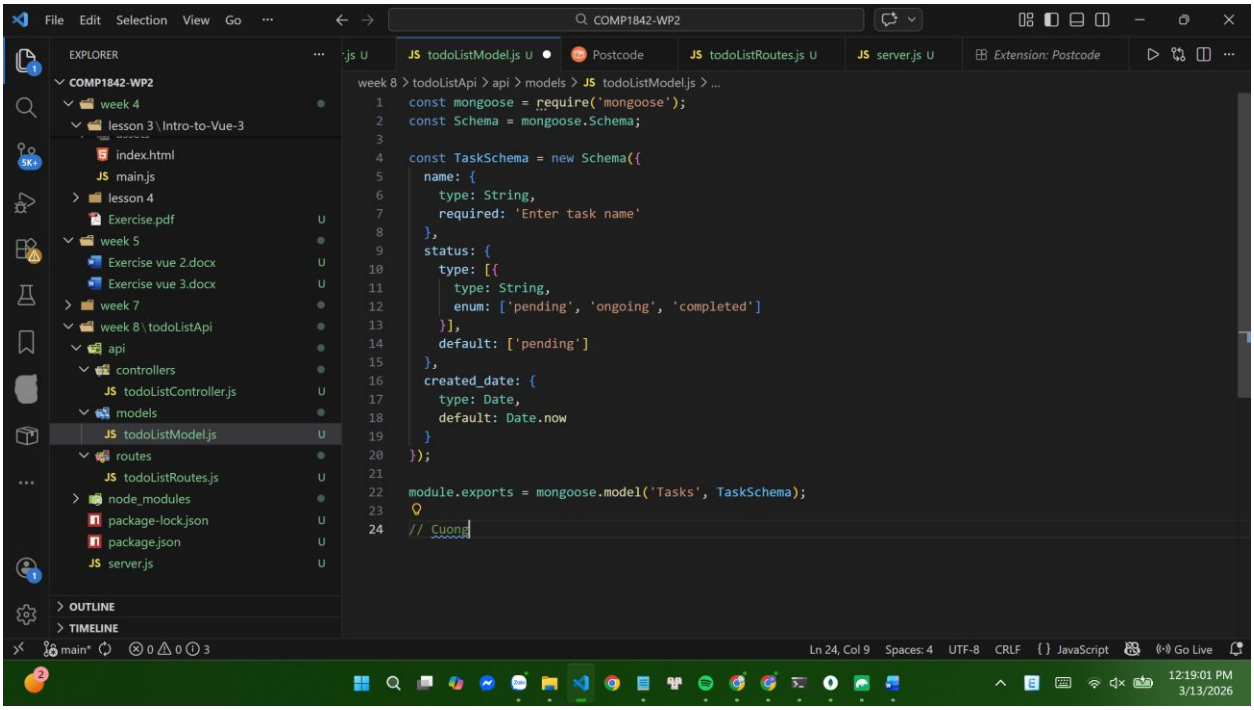
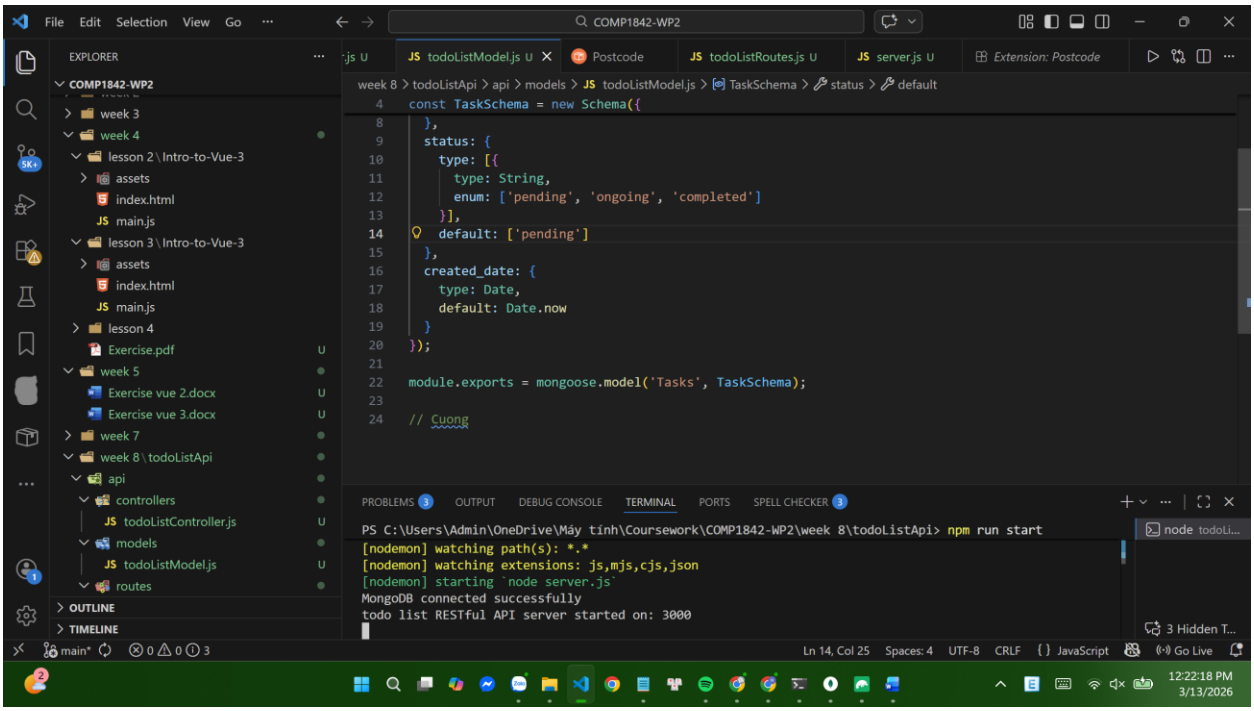
Week 8

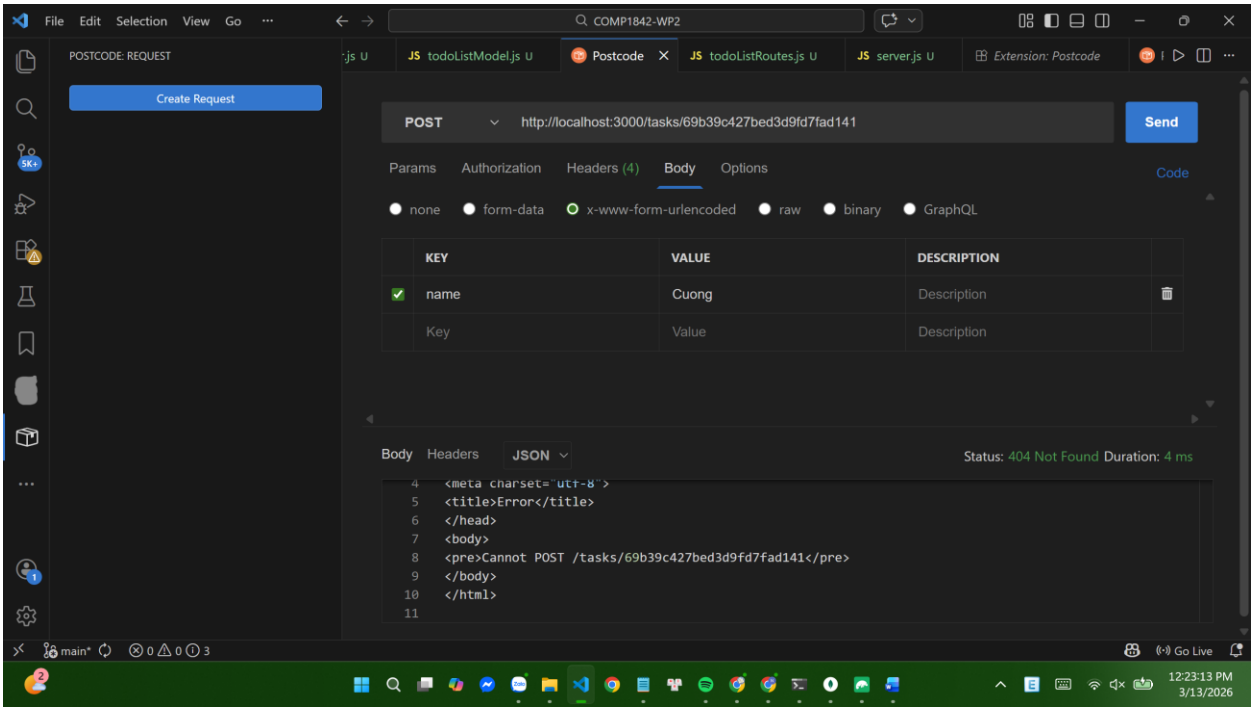
1. Screenshot 1:



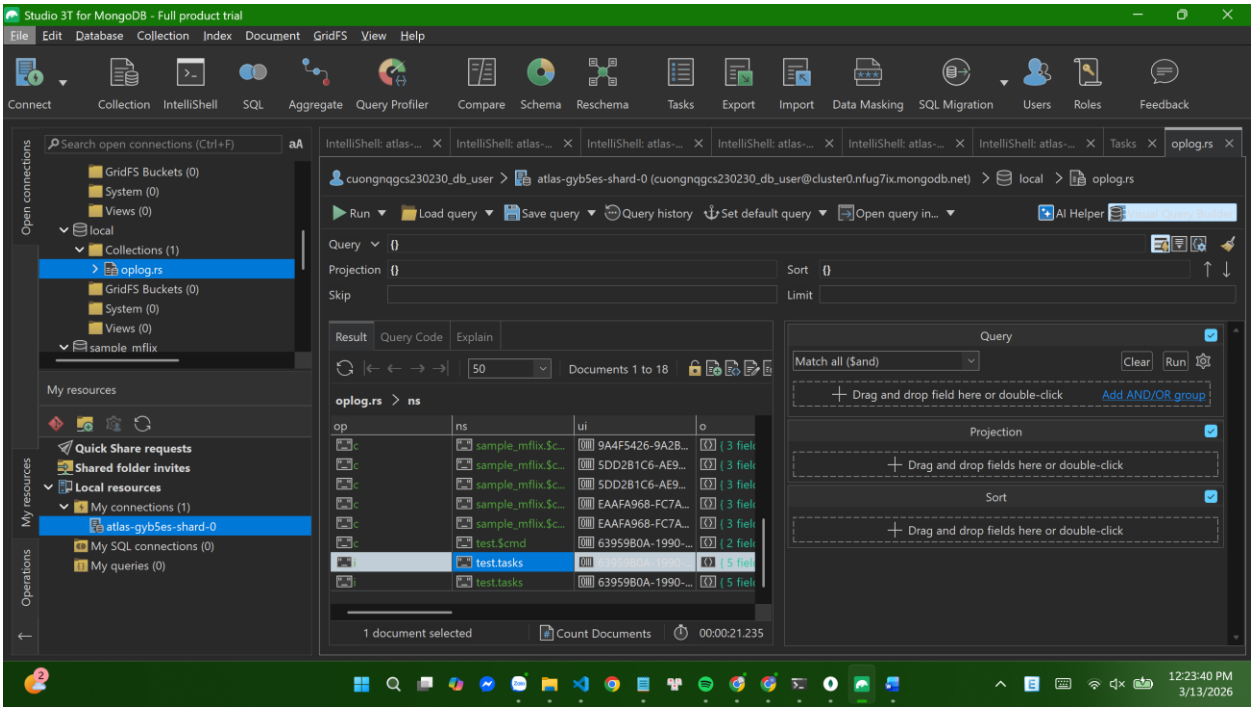
2. Screenshot 2:



3. Screenshot 3



4. Screenshot 4



5. Understanding part:
After studying and carefully reviewing the documentation on RESTful APIs, I gained a better understanding of how modern web applications communicate between clients and servers. An API (Application Programming Interface) is a mechanism that allows different software systems to communicate with each other using defined rules and

protocols. In a typical API architecture, the client sends a request to the server, and the server responds with the requested data. The client can be a web browser, a mobile application, or another software system that needs to access information from the server.

Another important concept I learned is REST, which stands for Representational State Transfer. REST is an architectural style that defines how web services are designed. RESTful APIs typically use standard HTTP methods such as GET, POST, PUT, and DELETE to perform operations on resources. For example, GET is used to retrieve data, POST is used to create new data, PUT is used to update existing data, and DELETE is used to delete data. These operations correspond to CRUD operations: Create, Read, Update, and Delete.

Another important part of the lecture was learning how to build a simple RESTful API using Node.js, Express, and MongoDB. The project structure divides the application into different components such as model, controller, and route. The model defines the data structures stored in the database, the controller handles the application logic, and the route defines how the API responds to different HTTP requests.

Finally, I also learned how to test API endpoints using tools like Postcode or other API testing tools. By sending requests like GET or POST to specific URLs, we can check if the server processes the request correctly and returns the expected JSON data. This practice exercise helped me understand both the theoretical concepts of RESTful APIs and how they are implemented in a real-world Node.js application.